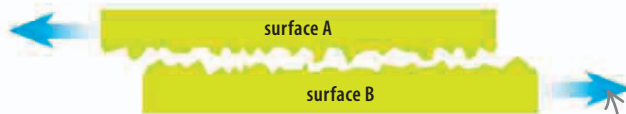


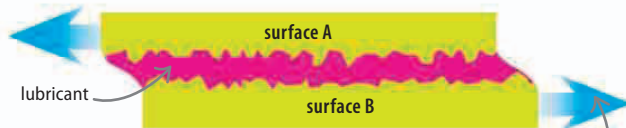
Friction and drag

Nothing in nature is perfectly smooth, so when objects slide past one another, their uneven surfaces push back against the direction of motion. This resistance force is known as friction. Drag is a similar phenomenon that occurs when an object pushes its way through air or water. The air or water pushes back, resisting the motion.



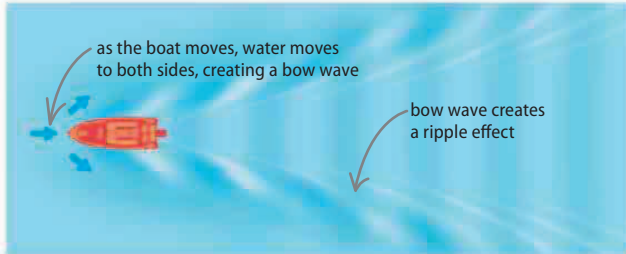
△ Friction

Even microscopic dips and bumps on a solid surface are enough to catch the uneven surface of another object moving over it, creating friction.



△ Lubrication

Adding a lubricant—generally a slippery liquid—reduces the friction. It provides a barrier that stops the solid surfaces touching as much.



△ Water resistance

The boat is moving through the water, and must push the water in front out of the way. The water resists and rises up as a bow wave.

REAL WORLD

Tire tread

Travel would be impossible without friction. For example, a tire has a rough tread to increase the friction force between the wheel and the road, preventing the car from sliding. Running shoes have rough soles for the same reason.



Resultant forces

Several different forces may act on one object at the same time, but sometimes the object cannot respond to each one individually. In these cases, the forces are combined to produce a single effect, so it appears that the object is being moved by a single force in one direction—this is the resultant force.

